iv) Sum of squares between blocks: $SSB = \sum_{j=1}^{m} \frac{x_{.j}^2}{k} - C.F$

v) Sum of Squares due to Error: $SSE = TSS\text{-}SST\text{-}SSB$

vi) Degrees of freedom

Degrees of freedom (d.f.)	d.f.
Total Sum of Squares	$n-1$
Treatment Sum of Squares	$k-1$
Block Sum of Squares	$m-1$
Error of Sum Squares	$(m-1)(k-1)$

vii) Mean Sum of Squares

Mean sum of Squares due to Treatments: $MST = \frac{SST}{k-1}$

Mean sum of Squares due to Blocks: $MSB = \frac{SSB}{m-1}$

Mean sum of Squares due to Error: $MSE = \frac{SSE}{(k-1)(m-1)}$

Step 5 : Calculation of the Test Statistic

ANOVA Table (two-way)

Source of variation	Sum of squares	Degrees of freedom	Mean sum of squares	F-ratio
Treatments	SST	k-1	MST	$F_{0t} = \frac{MST}{MSE}$
Blocks	SSB	m-1	MSB	$F_{0b} = \frac{MSB}{MSE}$
Error	SSE	$(k-1)(m-1)$	MSE	
Total	TSS	n-1		

Step 6 : Critical values

Critical value for treatments $= f_{(k-1,(m-1)(k-1)),\alpha}$

Critical value for blocks $= f_{(m-1,(m-1)(k-1)),\alpha}$

Step 7 : Decision

For Treatments: If the calculated F_{0t} value is greater than the corresponding critical value, then we reject the null hypothesis and conclude that there is significant difference among the treatment means, in atleast one pair.

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